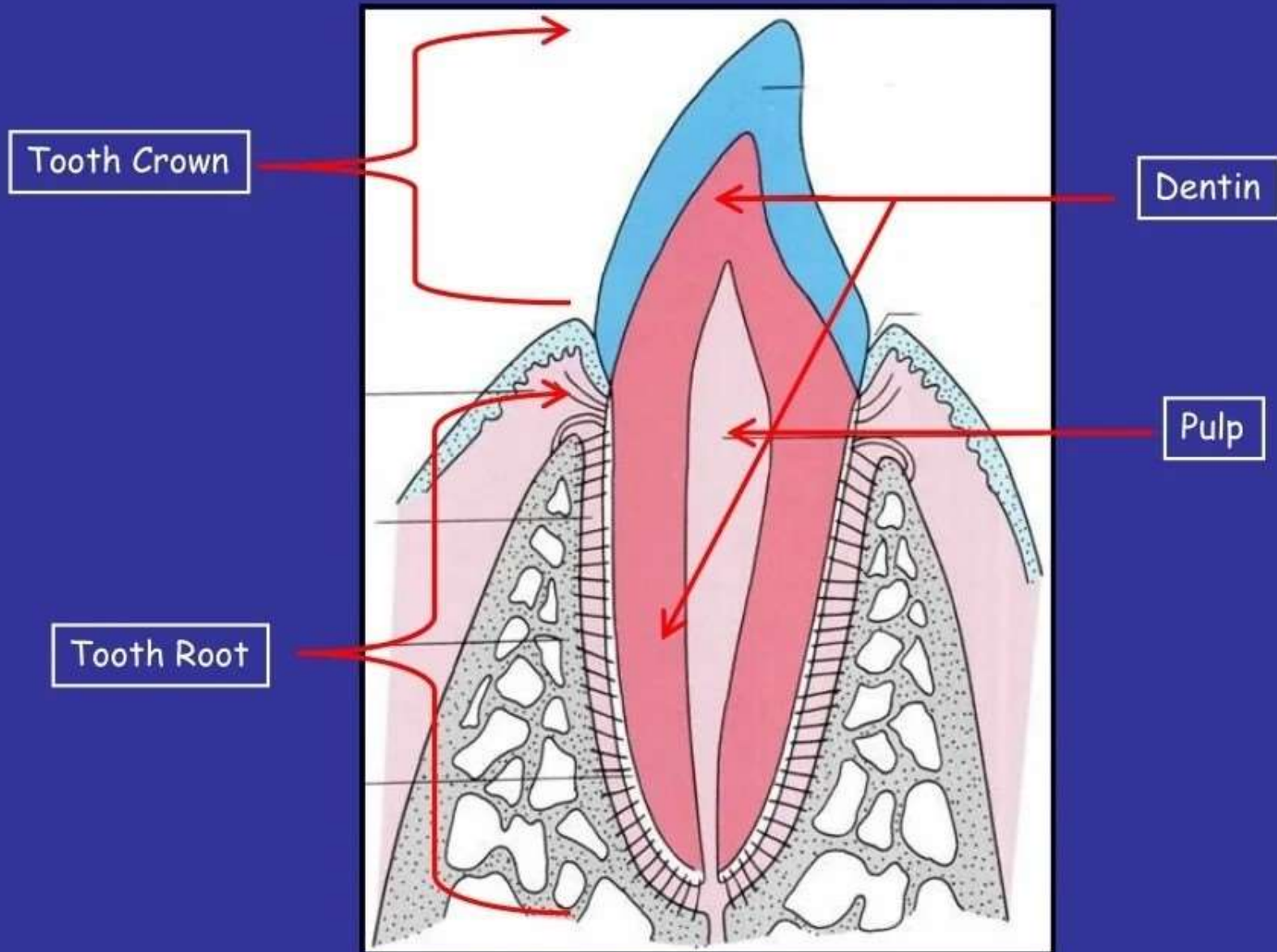
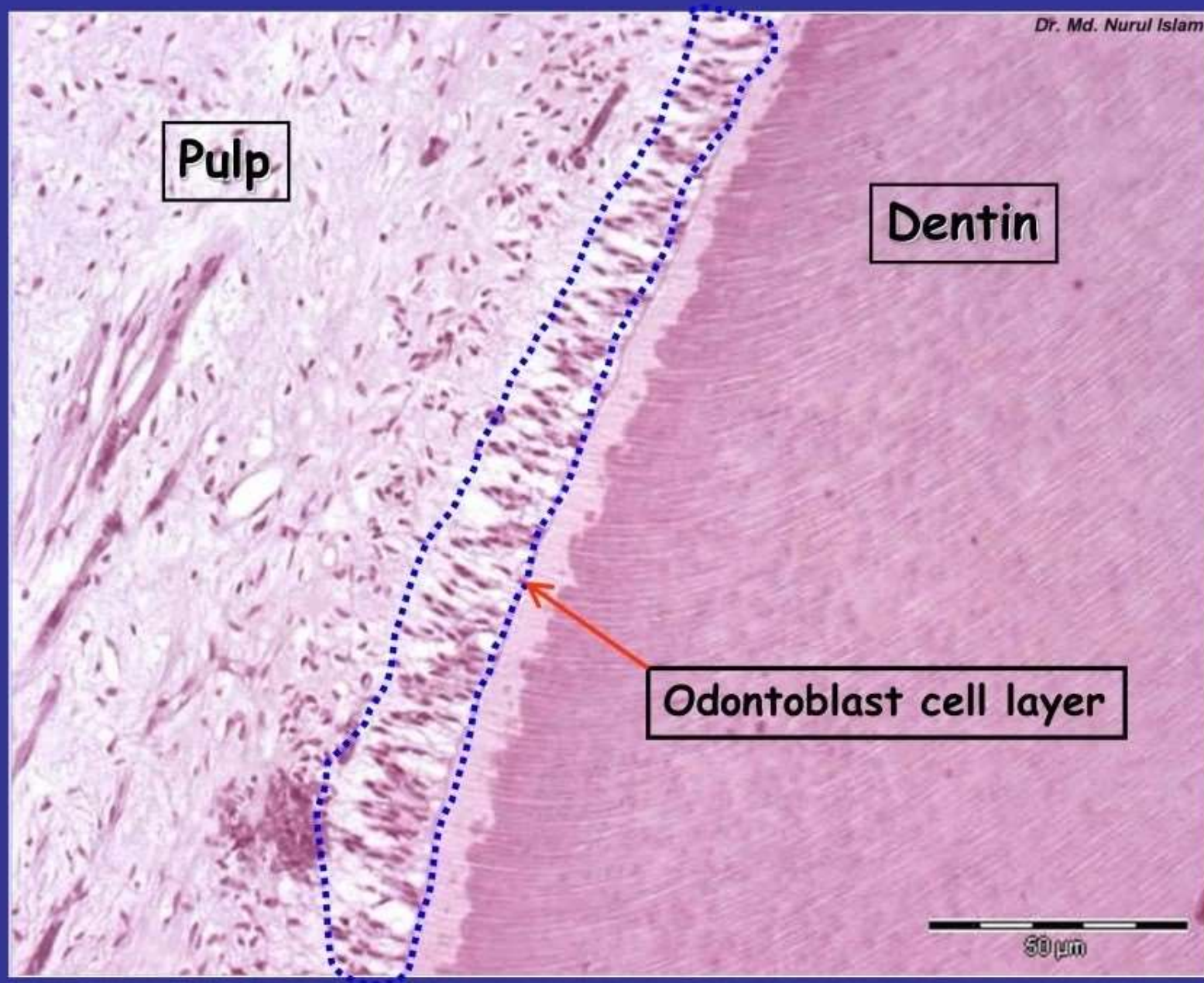


Dentin_pulp complex

Tooth





DENTIN

It is the mineralized tissue forming the main bulk of the tooth. ○

In the coronal part of the tooth, it is covered by the enamel and the radicular part by cementum. ○

Dentin is highly sensitive tissue. It is a vital tissue and it reacts to physiological stimulus. ○

Dentin composition:

70% inorganic hydroxyapatite crystals. ○

collagen). 10 I (Type substances organic 20% ○
%water.

This composition makes dentin more resilient than enamel. ○

DENTIN HISTOLOGICAL STRUCTURE

Dentin is composed of the following:

Dentinal tubules which contains .1
processes of odontoblasts.

Dentinal matrix. .2

The tubules follow an **S-shaped** curvature which ○
is more pronounced in crown and less in the root.

They are perpendicular to the DEJ & CEJ. ○

The tubules are larger in diameter near the pulpal ○
(4-3 μ m) and smaller at their outer ends

They contain the odontoblastic processes, ○
dentinal fluid and a transudate of plasma.

TYPES OF DENTIN (BASED ON TIME OF FORMATION)

Primary dentin: dentin formed before root completion and it outlines the pulp chamber. Include (Mantle dentin, circumpulpal dentin, peritubular dentin, and intertubular dentin).

Secondary dentin: dentin formed after root completion.

Tertiary dentin: it is formed in response to external stimuli such as dental caries, abrasion, attrition, trauma and some operative procedures.

Tertiary dentin is produced to protect the pulp from the stimulus. It is a **Defense reaction** against an injury. It is confined to the localized area of the pulp cavity wall.

It is deposited underneath the affected tubules.

This type of dentin has been subdivided ○ according to the presence or absence of the original odontoblasts.

If the original odontoblasts survive long enough to react to the dental caries, it produces dentin called **“Reactionary” dentin**.

If the odontoblasts are killed, new ○ odontoblast cells are differentiated from stem cells will produce dentin and it is called **“Reparative” dentin**.

CHANGES IN DENTIN WITH AGE:

Continuous formation of secondary dentin= reduction in size of pulp chamber (Pulp Recession).

Dentin sclerosis (calcification of dentinal tubules).

PULP

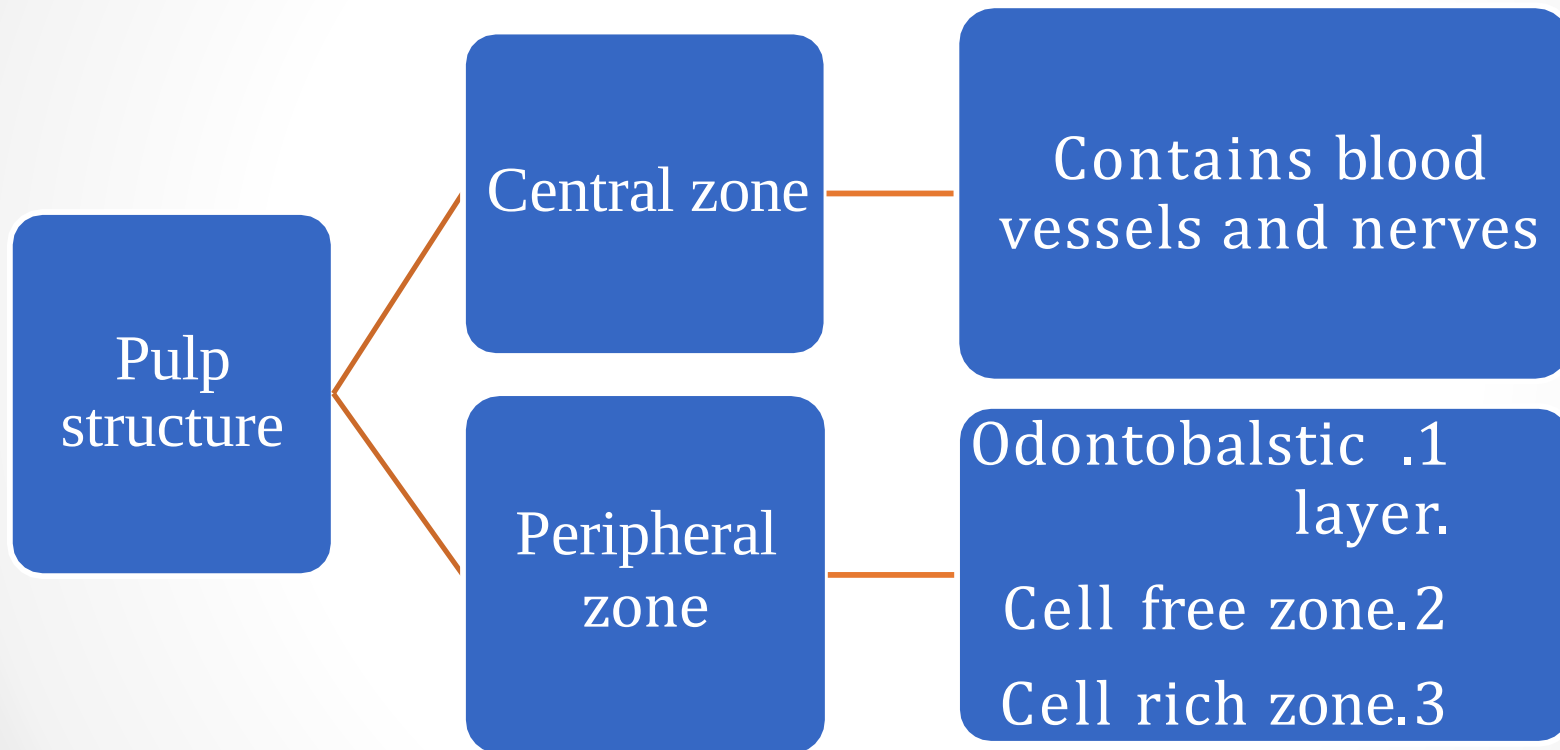
It is a soft connective tissue that forms the inner structure of a tooth. It contains cells, nerves and blood vessels.

The cells are odontoblast cells which are arranged peripherally in direct contact with the dentin.

Dentin-pulp complex

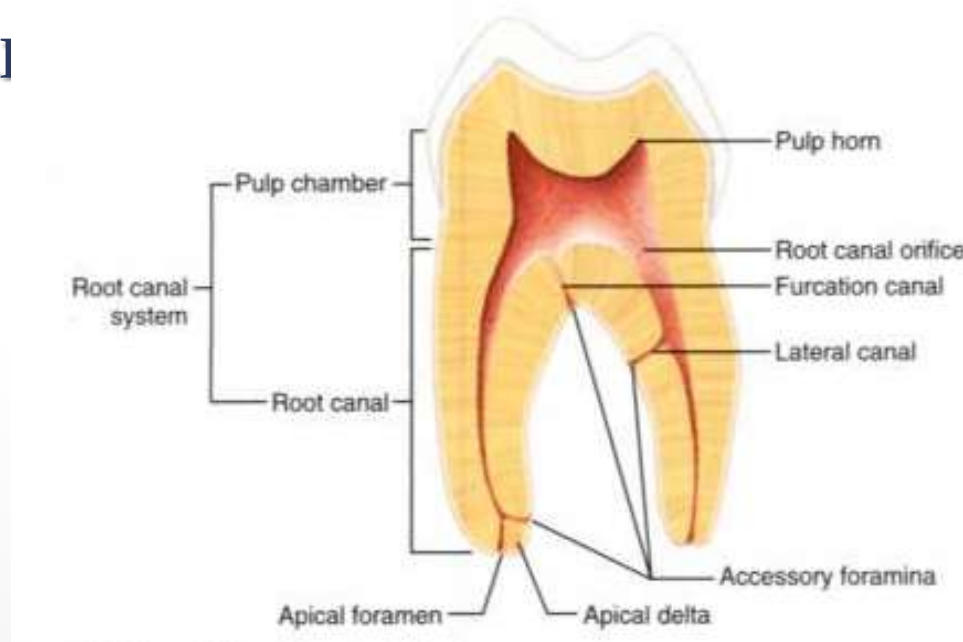
The pulp retains its ability to form dentin throughout life because of the presence of the undifferentiated mesenchymal cells and odontoblasts.

PULP HISTOLOGICAL STRUCTURE



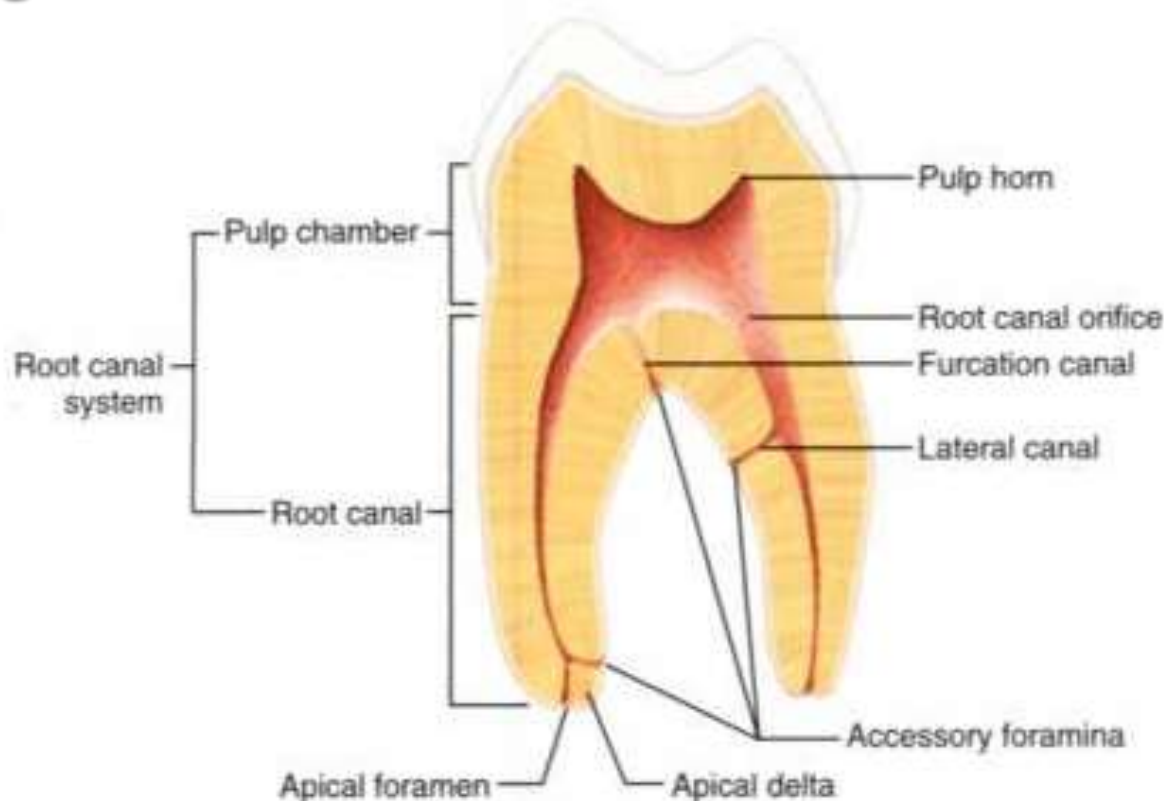
Pulp Cavity “Space”:

A central cavity within the tooth enclosed entirely by dentine except at the apical foramen. It is divided into coronal portions “pulp chamber” and a radicular



Pulp:

Is a loose connective tissue composed of blood vessels, nerves and cellular elements including odontoblasts which forms the dentine and surrounded by rigid mineralized dentine.



Coronal Pulp

Located coronally in the crown.

Has 6 surfaces (mesial, distal, buccal, lingual, floor).

Clinical Significant:

Care should be taken during cavity preparation to avoid exposure.

Age changes:

Secondary dentine, reparative dentine, pulp stones, size of the pulp chamber.

Radicular Pulp

That portion of the pulp that extends from canal orifice to the apical foramen.

Wide cervically and narrow near the apical foramen.

Varies in shape and number in different teeth.

Age changes:

The size of the root canal is greatest during *development & diminish gradually*.

Features of Pulp Which Distinguish it From Other Tissues

Pulp is surrounded by rigid walls so it is **unable to expand** ➤ in response to injury.

There is minimal collateral blood supply to pulp tissue ➤ which reduces its capacity for **repair**.

Pulp is connective tissue yet at its periphery are ➤ odontoblasts which deposit **secondary dentine**.

Innervation of pulp is **simple and complex**. ➤

Simple: contains only free nerve ending. ➤

Complex: cause high level of sensitivity to thermal and ➤ chemical change.

Types of Nerve fibers

Nerve fibers are classified according to their diameter, velocity of conduction and function to:

A fibers which have the largest diameter, (Lies superficially).
•1

C fibers which have the smallest diameter, (Deeply placed).
•2

A delta fibers are faster in conduction and are responsible for sharp localized dentinal pain.

C fibers are slower in conduction and are responsible for dull and throbbing pain.



Morphological changes

Continued deposition of intratubular dentine leads to reduction in the diameter of dentinal tubules.

Reduction in pulp volume due to increase in secondary dentine deposition. Root canal appears very thin or seem to totally obliterated.

Decrease in the number of pulp cells.

Degeneration and loss of myelinated and unmyelinated axons which lead to decrease in sensitivity.

Reduction in number of blood vessels.

Physiologic Changes

Decrease in dentine permeability provides protected environment for pulp reduced effect of irritants.

Possibility of reduced ability of pulp to react to irritants and repair itself.

Pulpal response to restorative procedures

During cavity and crown preparation pulp is subjected to many irritant

Thermal injury•

Vibration•

Desiccation of dentine•

Remaining dentine thickness•

Acid etching•

Prevention

Cutting procedure: use light, intermittent cutting, and efficient cooling system and high speeds of rotation.

Avoid desiccating the dentine: do not over dry the cavity preparation.

Do not apply irritating chemicals to freshly cut dentine.

Choose restorative material carefully (Biologically, physically).

Prevention

Do not use caustic cavity sterilizing agents.

Assume that all restorative materials will leak.

Use a cavity liner.

Do not use excessive force during insertion.

Employ polishing procedures that do not subject the pulp to excessive heat.

Establish a patient recall system that ensures periodic evaluation of the status of pulps that have been exposed to injury.

Vitality of pulp is important for health of teeth.

*Its sensory function has diagnostic value.
Its preservation and successful management
are two important challenges.*

Thank you